

[DC] The Portal as a Transition Between the Real and the Virtual

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Figure 1: Case-study 1: Glasshouse.

ABSTRACT

Our lives are increasingly shifting to the digital. Artists too have made attempts - successful or not - to attract audiences in virtual worlds. This evolution is irreversible, not to replace physical experiences, but to create new forms and tap into new audiences. To make these experiences valuable, it will be necessary to make all those involved - artists, performers, spectators - feel part of these virtual worlds. The portal, the access to this experience, plays a crucial role here. **How can the transition between the real and the virtual be constructed in such a way that those involved feel part of the virtual?** Concepts such as storytelling, interaction, presence and immersion are of great importance. Throughout my PhD research, I am conducting a series of experiments that combine these elements in various ways, aiming to develop portals in the form of performances and installations that bridge these two realms seamlessly.

Index terms: Virtual Reality, Immersive Storytelling, Presence, Sensory Involvement, Embodiment

1 INTRODUCTION

Unlike most artistic performances in traditional Western settings, virtual performances place the audience at the centre of the artistic creation - they put the spectator in the driver's seat for their own, unique experience. By putting the viewer at the centre of the narrative, Virtual Reality (VR) creators strive to create an environment where the distinction between observer and actor becomes blurred. In VR, the viewer is not merely an onlooker but an integral part of the unfolding story. The immersive nature of VR allows users to interact with and influence their surroundings,

forging a more profound connection with the content [7]. To make these experiences valuable, it is necessary to make all those involved - artists, performers, spectators - feel part of these virtual worlds. The "portal", the access to this experience, is hypothesized to play a crucial role here. **How can the transition between the real and the virtual be constructed in such a way that those involved feel part of the virtual?** This question lies at the heart of my PhD research (2022-2026), which recognizes the great importance of concepts such as storytelling, interaction, presence and sensory involvement to create engaging and embodied virtual experiences.

2 THE PORTAL AS A TRANSITION

A portal serves as a gateway that simplifies access to complex systems in information technology [6]. Similarly, in literature and film, it acts as a threshold enabling transitions between distinct realms—physical, virtual, or imagined—blending experiences and requiring active engagement. Portals embody transformation, connecting states of presence and perception. In science fiction, they symbolize gateways across spacetime, paralleling transitions between reality and virtuality.

This study draws on Lombard and Jones' model of spatial and social presence [4]. Spatial presence refers to "the subjective experience of being in one place or environment, even when one is physically in another" [4], while social presence involves "the feeling of being socially present with another person in a remote location" [4]. Milgram and Kishino's reality-virtuality continuum [5] further informs this exploration, bridging the physical and virtual as interconnected, rather than opposing, states. This continuum suggests infinite intermediate values, reflecting the nuanced relationship between reality and virtuality.

The philosophical concept of non-duality, as seen in Buddhism, resonates with these transitions, emphasizing the interplay between real, virtual, and latent spaces [1]. Hornecker highlights how VR engages the entire body, situating individuals in a space where the material and immaterial, analogue and digital, converge [3]. This embodied experience positions the audience within this dynamic continuum, reflecting the transformative potential of portals in immersive environments.

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3 CASE-STUDIES

Exactly that space in between, that transition an audience can experience when stepping into a virtual world, is the subject of my research. The portal serves as a metaphor for this transition, as an indispensable element to achieve a truly embodied feeling that goes far beyond just being surrounded by video. It is about being in different realities at the same time, it is about being confused, about losing your sense of space.

3.1 Glasshouse

The Glasshouse art installation (see [figure 1](#)) serves as a proof of concept, offering a multi-sensory, immersive experience that blurs the boundaries between physical and virtual realities. It combines a physical greenhouse filled with wild tomato plants and a bark-lined path with a VR experience using the Meta Quest 3. Visitors see their surroundings through the pass-through camera of the VR headset, which transitions into virtual visuals as they enter the greenhouse. This creates a seamless portal between the real and virtual worlds, with the visitor traveling from the greenhouse to a virtual rainforest in the Andes—the origin of the tomato plant. The installation highlights the history of the tomato and the environmental impact of modern greenhouse horticulture, particularly in Flanders and the Netherlands. By engaging multiple senses—sight, smell, touch, and sound—it creates a heightened sense of presence and embodiment. The concept draws on principles from the rubber hand experiment [2], which demonstrates how sensory integration can alter perceptions of body ownership. Similarly, the Glasshouse integrates physical and virtual stimuli through "touchpoints" along a 9-meter path, fostering a suspension of disbelief and a deeper connection to the experience. This innovative approach invites visitors to reflect on ecological and technological relationships in a multi-layered, immersive context.

3.2 Dissolution

Dissolution is a hybrid virtual reality experience combining theater and computer gaming for nine participants and one performer. It explores the theme of data consumption and digital waste through a narrative-driven, immersive performance. Participants start with a finite amount of data, and their decisions during the experience impact how quickly this data is consumed, ultimately determining the performance's duration.

The installation incorporates physical elements to enhance presence, with transparent cube-shaped structures functioning as portals between the real and virtual. These touchpoints reinforce the connection between physical and digital realms. The experience critiques modern data consumption practices while immersing participants in a world where every action has tangible consequences on shared resources.



Figure 2: Case-study 2: Dissolution

Building on insights from the earlier Glasshouse case study, Dissolution introduces new elements to expand its immersive scope. Unlike Glasshouse, which was a single-person experience, Dissolution allows up to 9 participants to enter the virtual space simultaneously. This shared experience fosters a deeper sense of social presence, as spectators can encounter and interact with each other within the virtual environment. Additionally, the narrative in Dissolution is more developed, enhancing emotional involvement and engagement.

4 CONCLUSION

The projects Glasshouse and Dissolution illustrate my ongoing exploration of the interplay between physical and virtual spaces, as well as the potential of multi-sensory experiences to deepen immersion and engagement.

While these case studies provide a foundation for my research, key challenges and opportunities for development remain. First, as my work lies at the intersection of art and research, I am seeking guidance on how to make my methodology and findings more scientific without losing the artistic essence of the inquiry. How can I effectively balance rigorous scientific methods with the creative and exploratory nature of PhD research in the arts?

Second, as I plan the next case study, I aim to build on the lessons learned from Glasshouse and Dissolution. I would like to discuss strategies for designing a new case study that both refines my research questions and provides meaningful insights into the broader field of immersive art. Specifically, how can I further develop and measure immersion and social presence in a way that contributes to both artistic practice and academic discourse?

Lastly, I welcome feedback on how to structure my research to foster interdisciplinary collaboration while ensuring it remains grounded in the conceptual framework of my dissertation. These discussions will be invaluable in shaping the next phase of my PhD journey and strengthening the contribution of my work to the field of immersive experiences.

FIGURE CREDITS

[Figure 1](#) image credit: Wannes Cré.

[Figure 2](#) image credit: Kristof Timmerman.

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